Benchmarking Neural Surrogates for Hydrofoil RANS Flow, Cavitation-Risk Prediction, and Hydrodynamic Shape Optimization

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Abstract—We benchmark four neural surrogates for hydrofoil RANS prediction and design: CNN-U-Net, a supervised physics-regularized point network, a Fourier neural operator (FNO), and DeepONet. The dataset contains 381 valid OpenFOAM cases. Its 64-case validation set consists entirely of two NACA families that are absent from training. The models predict flow fields, force coefficients, and pressure-based cavitation risk. The physics-regularized network gives the best pressure result ($R^2 = 0.966$) and cavitation-risk F1 score (0.921). DeepONet gives the best lift and drag results ($R^2 = 0.996$ and 0.988). Inference is 917–1801 times faster than OpenFOAM. We also use each surrogate for continuous NACA shape optimization and rerun its best design in OpenFOAM. The resulting lift-to-drag improvements range from 2.3% to 79.7% relative to the starting foils. This end-to-end benchmark compares accuracy, speed, and the quality of the final CFD-validated designs.

Index Terms—AI for science, CFD surrogate, neural operator, physics-informed learning, hydrofoil optimization, cavitation inception

I. Introduction

Computational fluid dynamics (CFD) is widely used for hydrofoil analysis and design. The difficulty is cost: optimization and operating sweeps may require hundreds or thousands of Reynolds-averaged Navier–Stokes (RANS) solutions. Neural surrogates reduce that repeated cost and are increasingly useful in data-driven fluid mechanics [1]. Their value should be measured beyond field error. A model must also preserve forces, identify low-pressure regions, and guide an optimizer toward a design that remains good when it is returned to CFD.

CNN-U-Net, FNO, DeepONet, and physics-regularized networks represent the flow map in different ways [2], [3], [4], [5]. That makes them a useful benchmark set. Prior work often compares field metrics alone. We extend the comparison to pressure, lift, drag, cavitation-risk classification, runtime, and hydrodynamic shape optimization.

Earlier airfoil surrogates used convolutional or hybrid networks to predict steady flow fields from geometry and operating conditions [6], [7], [8], [9]. Physics-informed neural operators provide another route for combining PDE structure with operator learning [10]. Our study instead holds the data and downstream optimizer fixed to compare four model families directly.

Twelve NACA four-digit families, eight angles of attack, and four Reynolds numbers yield 384 OpenFOAM cases. After a force-coefficient quality check, 381 cases remain. The validation set contains every case from NACA 0015 and 2412, so it measures generalization to geometries not used for training. All four models use the same targets and split. Each model is then placed in the same optimizer, and its selected design is independently rerun in OpenFOAM.

The contributions are:

- a four-model benchmark on an unseen-geometry validation set, covering flow fields, force coefficients, runtime, and cavitation-risk screening;
- a common hydrodynamic shape-optimization experiment for all four surrogates; and
- an independent OpenFOAM rerun of each model's selected design.

Here, "cavitation risk" means pressure-threshold inception risk computed from C_p. It is useful for screening low-pressure regions. It is not a multiphase prediction of cavity length or shedding.

II. Data and CFD Reference

A. Design space

The dataset contains NACA 0006, 0008, 0010, 0012, 0015, 0018, 2412, 2415, 4412, 4415, 4418, and 4421 profiles generated from the standard four-digit parameterization [11]. For each profile, we simulate angles of attack $\alpha \in \{-4, -2, 0, 2, 4, 6, 8, 10\}^\circ$ and Reynolds numbers $Re \in \{10^5, 2\times10^5, 5\times10^5, 10^6\}$. The chord is 1 m, water density is 997 kg/m^3, and kinematic viscosity is 10^{-6} m^2/s, so the freestream speed is set by $U_\infty = Re\nu/c$. The far-field absolute pressure and vapor pressure are 101325 and 2300 Pa.

Each case is solved with OpenFOAM simpleFoam [12], using steady incompressible RANS and the k–ω SST closure [13]. Angled freestream velocity is imposed at the inlet and far boundary. The solver runs for 300 iterations, and OpenFOAM's forceCoeffs provides total lift and drag references. Cell-centered fields are interpolated to a common 128×64 Cartesian grid over $x/c \in [-1, 2]$ and $y/c \in [-0.75, 0.75]$. Inputs include coordinates, operating parameters, three NACA geometry descriptors, a fluid mask, and signed distance to the profile.

An audit found no pressure fields duplicated across angle of attack and found force files for all 384 cases. Three cases produced $|C_L|$ or $|C_D| > 5$ and were removed before splitting. This leaves 381 cases. All 64 cases from NACA 0015 and NACA 2412 form the unseen-geometry validation set. The remaining 317 cases train the models. No validation geometry therefore appears in training at another operating condition.

B. Targets and cavitation-risk definition

The common output vector contains U_x, U_y, absolute pressure p, eddy viscosity, k, ω, C_p, cavitation margin, C_L, and C_D. Grid fields use masked normalized mean-squared error, while the case-level force targets use normalized mean-squared error. OpenFOAM's kinematic gauge pressure p_k is converted to absolute pressure as

$$p = p_\infty + \rho p_k, \qquad C_p = \frac{p - p_\infty}{\frac{1}{2}\rho U_\infty^2}. \tag{1}$$

For an ambient-pressure value p_a, the pressure-based inception margin is

$$M(x, y; p_a) = p_a + C_p(x, y)\tfrac{1}{2}\rho U_\infty^2 - p_v. \tag{2}$$

A cell is labeled at risk when $M < 0$. We evaluate this label at $p_a \in \{2500, 3000, 5000, 10000, 25000, 101325\}$ Pa. This sweep creates a useful classification test from the predicted C_p field without representing multiphase physics.

III. Surrogates and Optimization

A. Model families

The CNN-U-Net has three encoder/decoder levels, skip connections, group normalization, GELU activations, and base width 16. It maps the complete input grid to the output grid [2]. The FNO has width 32, four operator blocks, and 12 retained Fourier modes in each direction. One block is

$$h_{\ell+1} = \sigma\big(W_\ell h_\ell + \mathcal{F}^{-1}(R_\ell \mathcal{F}(h_\ell))\big). \tag{3}$$

following the spectral operator construction of [3]. Here, W_ℓ is a local linear map and R_ℓ acts on the retained spectral modes. The DeepONet uses three-layer, width-64 branch and trunk networks with 64 basis functions. For output k,

$$\widehat{y}_k(\mathbf{x}; \boldsymbol{\mu}) = \frac{\mathbf{b}_k(\boldsymbol{\mu})^\top \mathbf{t}_k(\mathbf{x})}{\sqrt{64}} + c_k, \tag{4}$$

as in the branch–trunk construction of [4], where $\boldsymbol{\mu}$ contains the operating and geometry parameters.

The physics-regularized MLP has three tanh layers of width 64. It is supervised by CFD values at 1,024 sampled points per case. Its extra loss uses the directional residuals

$$r_c = u_x + v_y, \quad r_x = uu_x + p_x/\rho, \quad r_y = vv_y + p_y/\rho, \tag{5}$$

with $\mathcal{L} = \mathcal{L}_{\text{sup}} + 10^{-4}\mathcal{L}_{\text{phys}}$, following the residual-loss principle in [5]. This is a lightweight physics regularizer, not a data-free PINN or a complete RANS residual.

All models use AdamW with learning rate 2×10^{-3}, weight decay 10^{-4}, at most 120 epochs, patience 20, gradient clipping, and seed 7. Normalization is computed from training cases only. The best checkpoint is selected by validation loss. Model size ranges from 9,930 to 595,914 parameters (Table I).

Figure 1 reports the complete optimization histories rather than only the selected checkpoints. Because the architectures consume the data in different forms and the objective aggregates heterogeneous normalized targets, loss magnitude should be interpreted within a model, not as a direct cross-model accuracy metric. DeepONet reaches its minimum validation objective at epoch 6, the physics-regularized network at epoch 30, FNO at epoch 54, and CNN-U-Net at epoch 120. The isolated FNO spikes are retained in the record and motivate reporting best-checkpoint results alongside the full history.

Although surface-pressure integration was implemented as a diagnostic, the coarse common grid did not reproduce absolute OpenFOAM forces reliably because it omits viscous traction and under-resolves the wall. Therefore, optimization uses dedicated C_L and C_D output heads trained on OpenFOAM's total force coefficients. This choice avoids presenting an invalid pressure-only force calculator as ground truth, but it also means force prediction is directly supervised.

B. Common optimization protocol

At $Re = 5 \times 10^5$, every model optimizes the NACA variables $\theta = (m, p, t, \alpha)$ from NACA 0012, 2412, and 4415 starts. Bounds are $m \in [0, 0.09]$, $p \in [0.1, 0.9]$, $t \in [0.06, 0.20]$, and $\alpha \in [-4, 10]^\circ$. Nelder–Mead [14] runs for at most 40 iterations with identical tolerances. The objective is

$$J(\theta) = \frac{C_L}{|C_D| + 10^{-6}} - 10P_M - 5f_M, \tag{6}$$

where P_M penalizes a predicted minimum cavitation margin below 5000 Pa and f_M is the fraction of cells below that margin. At the nominal ambient pressure, all selected designs remain far above the threshold, so the cavitation penalty is inactive; the separate ambient-pressure sweep evaluates inception classification under reduced pressure. For each architecture, we select the highest-scoring design across starts and create a new mesh and OpenFOAM case without using surrogate predictions to initialize the solver.

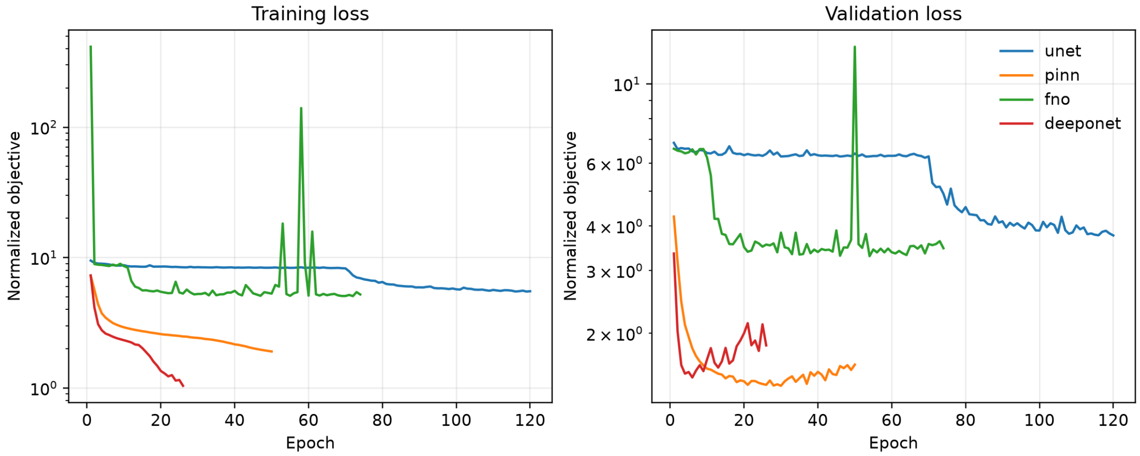

Fig. 1. Training and unseen-geometry validation objectives. Curves terminate after early stopping; objectives combine normalized field and force losses.

TABLE I
Validation performance, model size, and cost. Force RMSE is computed case-wise against OpenFOAM total coefficients.

Model	p RMSE	p R^2	C_L RMSE	C_D RMSE	Params.	Speedup
CNN-U-Net	28.52	0.392	0.0991	0.00453	485,178	1,148×
Physics-reg. MLP	6.78	0.966	0.0245	0.00204	9,930	1,801×
FNO	22.73	0.614	0.1383	0.00957	595,914	1,028×
DeepONet	8.55	0.945	0.0268	0.00184	100,746	917×

IV. Results

A. Validation accuracy and computational cost

Table I reports results on the 64-case validation set. The physics-regularized network is best for pressure and C_p, with a pressure RMSE of 6.78 Pa and $R^2 = 0.966$. DeepONet gives the best U_x, U_y, C_L, and C_D coefficients. The grid operators are weaker in this low-data setting. FNO retains useful force trends. CNN-U-Net predicts C_p and forces more accurately than absolute pressure or streamwise velocity. The normalized multi-target loss and the affine relationship between p and C_p allow this difference between channels.

Measured training times are 868, 104, 547, and 54 s for CNN-U-Net, the physics-regularized network, FNO, and DeepONet, respectively. Median OpenFOAM execution time is 12.4 s per case, while pressure-field inference requires 0.0069–0.0135 s, yielding 917–1801× speedups. The benchmark used an 8-core Apple M2 Mac mini with 16 GB memory; PyTorch 2.13 ran on CPU and single-process OpenFOAM 9 ran in Docker. These numbers measure repeated local inference and exclude data generation and mesh construction.

Figure 3 shows a representative validation case from the unseen NACA 0015 family. The pointwise models reproduce the leading-edge pressure structure most closely; CNN-U-Net smooths the extrema and FNO introduces spatial texture. The distinct error color scales in the lower row are retained to expose each model's error structure rather than visually equalizing models with very different ranges.

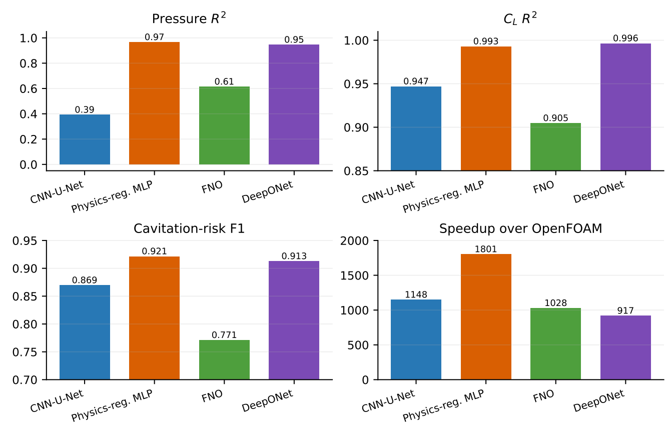

Fig. 2. No architecture dominates every criterion on the 64-case validation set. Cavitation F1 is aggregated over ambient-pressure sweeps.

B. Cavitation-inception screening

The pressure-threshold labels are highly imbalanced, so accuracy is not useful here. The physics-regularized network achieves precision 0.927, recall 0.916, and F1 0.921. DeepONet follows at F1 0.913. CNN-U-Net has the highest recall (0.958) but lower precision (0.796), while FNO obtains F1 0.771. Twenty-four validation case/ambient-pressure combinations contain at-risk cells. This test measures whether a surrogate preserves low-C_p regions well enough for inception screening. It does not model cavity dynamics. That would require multiphase RANS labels such as vapor volume fraction and cavity length [15].

C. CFD-revalidated optimization

Table II and Fig. 4 compare each model's predicted winner with a fresh OpenFOAM solve. All four winners

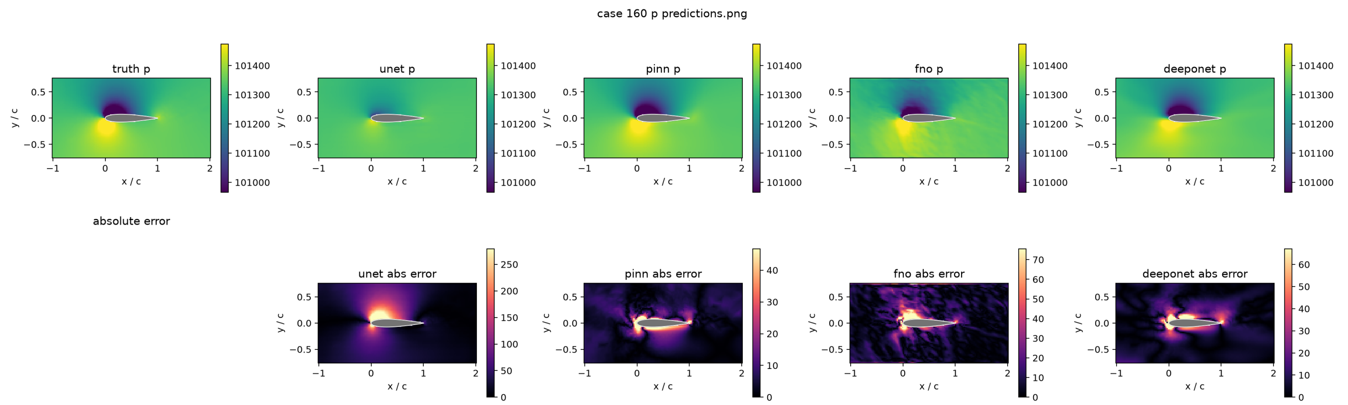

Fig. 3. Absolute-pressure prediction for a representative validation case from the NACA 0015 case (case 160), followed by per-model absolute error. All prediction panels share the truth color scale; error panels use individually labeled scales.

TABLE II
Surrogate-selected winners and independent OpenFOAM revalidation. Improvement is relative to the corresponding start at the optimization condition.

Model	Start	Pred. L/D	CFD L/D	Error	CFD gain
CNN-U-Net	2412	30.17	38.45	−21.5%	48.5%
Physics-reg. MLP	4415	43.62	45.96	−5.1%	79.7%
FNO	4415	21.11	26.15	−19.3%	2.3%
DeepONet	2412	39.99	40.86	−2.1%	57.8%

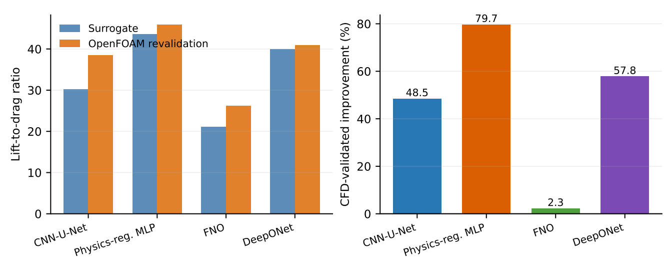

Fig. 4. Predicted and OpenFOAM-revalidated performance of each surrogate's selected design. Error in absolute L/D does not by itself determine whether a useful improvement is found.

improve upon their associated starting foil in CFD, but by very different amounts. The physics-regularized network selects a thin, strongly cambered profile whose validated $L/D = 45.96$ is 79.7% above the NACA 4415 baseline. DeepONet obtains a 57.8% improvement and CNN-U-Net obtains 48.5%. FNO predicts the lowest-performing winner; revalidation gives only a 2.3% gain over its baseline.

The optimization test sharpens the field benchmark in two ways. First, DeepONet gives the smallest winner-level L/D discrepancy, consistent with its force accuracy, but the physics-regularized model finds the best validated design. Second, the FNO winner improves after CFD revalidation yet remains close to its baseline, showing that a conservative prediction error does not guarantee strong search guidance. Screening of five randomly perturbed profiles and an oval was also performed, but these geometries lie outside the NACA training manifold and are reported only as stress tests, not validated free-form design evidence.

V. Scope and Next Steps

The pointwise models outperform the grid operators in this low-data, fixed-grid experiment. The result is useful precisely because all four models see the same cases, targets, split, and optimization task. It is not a universal ranking. Larger operator-learning datasets or different loss balancing may favor CNN-U-Net or FNO.

Pressure accuracy, force accuracy, and optimization quality capture different parts of the workflow. DeepONet is the most accurate force model, while the physics-regularized network returns the best CFD-validated design. The optimizer concentrates queries in a small region and can amplify structured prediction error. For that reason, a CFD solve remains the final step in the design loop.

The current scope is two-dimensional, steady, single-phase RANS over NACA four-digit profiles. Three divergent force cases are removed by a fixed quality rule. A symmetry audit also finds a small lift bias at zero incidence. The study does not yet include grid-convergence or turbulence-model sensitivity, and the validation families do not represent arbitrary topology. Force coefficients are directly supervised because the common grid cannot resolve pressure and viscous traction accurately enough for force integration. We rerun one winner per model. Cavitation is evaluated as pressure-threshold inception risk, not vapor transport or unsteady shedding.

The next steps are repeated training seeds, mesh and turbulence sensitivity, more CFD checks around each optimum, uncertainty-guided sampling, and multiphase or experimental cavitation labels. These extensions broaden the claim; they do not change the present benchmark result that architecture choice affects both prediction accuracy and the design returned by the optimizer.

A. Practical model selection

The timing results are most useful when interpreted as amortized workflow costs. Dividing each measured training time by the per-case time saved relative to OpenFOAM gives approximate break-even points of 70 new evaluations for CNN-U-Net, 8 for the physics-regularized network, 44 for FNO, and 5 for DeepONet. These values exclude the CFD corpus, which is a shared prerequisite for all four supervised models, but they show that training cost is recovered quickly once a model is used for optimization or a broad operating sweep. The physics-regularized model is especially favorable here: its parameter count is roughly 1/49 that of CNN-U-Net and 1/60 that of FNO, yet it has the fastest inference and strongest validated design result.

Speed alone, however, is not an adequate deployment rule. A practical workflow can separate three roles. First, a fast surrogate screens thousands of candidate operating points or geometries. Second, a verification policy sends selected candidates back to CFD based on predicted merit, model disagreement, distance from the training manifold, or uncertainty. Third, verified cases are appended to the training set before another optimization round. The present experiment implements the first role and a fixed version of the second by revalidating one winner per architecture. It does not yet adaptively retrain. The disagreement between FNO's search outcome and those of the pointwise models suggests that cross-model consensus could already serve as a simple warning signal: a design preferred only by the weakest model on the validation set deserves a larger verification budget than one supported by several independently structured surrogates.

Architecture selection should therefore follow the intended scientific action. DeepONet is attractive when accurate force coefficients and faithful ranking are primary. The physics-regularized network is the strongest choice in this low-data study when pressure, inception screening, runtime, and final design improvement are considered jointly. CNN-U-Net's high inception recall may be useful for conservative screening, despite its poorer pressure field, while the current FNO requires more data or tuning before design deployment. This task-specific view is more actionable than naming a single winner from an averaged multi-output loss.

B. Reproducibility and evidence boundaries

The benchmark was built around machine-readable provenance rather than a single summary spreadsheet. Each processed case retains its operating parameters, geometry descriptors, source identifier, fluid mask, signed distance, field targets, and OpenFOAM force coefficients. The deterministic split is regenerated from the valid case list and seed, while every checkpoint stores its target and input field order, normalization, architecture arguments, and training history. Evaluation scripts write per-case CSV files before producing aggregate metrics. Similarly, optimization records every parameter vector and objective evaluation, and the four revalidation directories contain the generated meshes, solver logs, final fields, and force histories. This structure permits headline values to be traced from a table back to the underlying case rather than reconstructed from a plot.

The audit also records exclusions and failed checks. Exact cross-angle pressure duplicates are absent in the corrected dataset, three nonphysical force cases are excluded by a predeclared $|C| \leq 5$ rule, and random or oval profiles are marked as tests outside the training geometry family. We report the small symmetric-foil lift bias rather than using symmetry as an assumed validation. For runtime, the scripts parse OpenFOAM's logged execution time and time pressure-field inference over the same selected cases; they do not fold the one-time training cost into the speedup. For cavitation, the evaluation artifact labels every result as pressure-threshold inception risk. These boundaries are part of the result: an auditable scientific ML workflow should preserve not only successful metrics but also exclusions, inactive constraints, and conditions under which a claim ceases to apply.

VI. Conclusion

We benchmarked four neural surrogates for hydrofoil RANS prediction, cavitation-risk screening, and shape optimization. The physics-regularized network gives the best pressure accuracy, inception-risk F1, speed, and CFD-validated design improvement. DeepONet predicts force coefficients most accurately. CNN-U-Net remains competitive for C_p, force prediction, and high-recall risk screening, while FNO is weaker in this low-data setting. All models are 917–1801× faster than the measured OpenFOAM solves. The optimization results show why a useful surrogate benchmark should include both validation accuracy and CFD checks of the designs the models select.

Artifact Availability

Code, configurations, split definitions, trained checkpoints, per-case metrics, optimization traces, and OpenFOAM validation cases are organized in an auditable repository. An anonymized archival link will be added before submission.

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